

A PROOFS

A.1 Theorem 1

PROOF. We assume that there is a possible solution. Given that $\mathcal{T}_i$ and $\mathcal{S}_i$ are executable within polynomial time, we can also verify whether the solution is correct or not in polynomial time. Thus, the problem is in NP. To prove the NP-complete, we show a reduction from the decision problem to the set cover problem, one of Karp's 21 NP-Complete problems [19].

We construct an instance of repair problem as follows. First, we denote $r_j = x_j - t_j - s_j$. Let $n = \min(p, q)$, $m = 2$, $\eta = n$, $x_i = 0$ for $1 \leq i \leq n$. For trend filters $\mathcal{T}_i$ and seasonality filters $\mathcal{S}_i$, $t_{j+1} = \mathcal{T}_{j+1}(x_{j+1}, x_j) = x_{j+1} - x_j$ if $u_i \in Y_j$ and $(i - j) \nmid 2$ otherwise 0. $s_i = \mathcal{S}_i(\{d_j; 1 \leq j \leq n \wedge (j - i) \mid m\}) = -n - 1 + 2 \sum_{u_i \in Y_j, (i-j) \mid 2} (x_j - t_j) + 2 \sum_{u_i \in Y_j, (i-j) \nmid 2} (x_{j+1} - t_{j+1})$ for $1 \leq i \leq n$. If $p > q$, we add some artificial x_i, t_i, s_i where $x_i = 0, t_i = \mathcal{T}_i(x_i, x_{i-1}) = 0, t_i = \mathcal{S}_i(\{d_j; 1 \leq j \leq n \wedge (j - i) \mid 2\}) = 0$ for $n < i \leq p$, i.e., $r_i = 0 \leq n$ is always satisfied. If $p < q$, we add some artificial x_i where $x_i = 0$ for $n < i \leq q$. Finally, we set the repaired value of x_i , i.e., $x'_i = 1$. Figure 3 shows an example. Given $Y_3 = \{u_1, u_6\}$, $p = 6$, $q = 7$, we construct an instance with $n = \min(p, q) = 6$, $x_i = 0$ for $1 \leq i \leq 7$. Since $u_6 \in Y_3$ and $(6-3) \nmid 2$, $t_4 = \mathcal{T}_4(x_3, x_4) = x_4 - x_3$. Otherwise $t_i = 0$ for $i \in \{1, 2, 3, 5, 6\}$. $s_1 = \mathcal{S}_1(x_1 - t_1, x_3 - t_3, x_5 - t_5) = -n - 1 + 2(x_3 - t_3) = -n - 1 + 2x_3$. $s_6 = \mathcal{S}_6(x_2 - t_2, x_4 - t_4, x_6 - t_6) = -n - 1 + 2(x_4 - t_4) = -n - 1 + 2x_3$. Otherwise $s_i = -n + 1$ for $i \in \{2, 3, 4, 5\}$. We can derive that $r_1 = x_1 - t_1 - s_1 = n + 1 + x_1 - 2x_3$, $r_6 = x_6 - t_6 - s_6 = n + 1 + x_6 - 2x_3$, $r_4 = n + 1 + x_3$, $r_1 = r_2 = r_5 = n + 1$. Therefore, when $x_3 = 1$ in the gray box, $r_1 = r_6 = n - 1 \leq \eta$ in the blue box, while $r_4 = n + 2 > \eta$, $r_1 = r_2 = r_5 = n + 1 > \eta$. It is clear that we can construct $x_i, \eta, \mathcal{T}_i, \mathcal{S}_i$ in polynomial time.

We now show that $\Delta(x, x') \leq \tau$ if and only if $\text{card}(L) \leq \tau$. Begin by assuming $\text{card}(L) \leq \tau$ has a solution L , repair x_i to $x'_i = 1$ for each $Y_i \in L$. Thereafter, for $u_j \in Y_i$, it is observed that $-n + 1 \leq s_j \leq n - 1$. If $u_i \in Y_{j-1}$ and $(i - j) \nmid 2$, $t_j = \mathcal{T}_j(x_j, x_{j-1}) = x_j - x_{j-1}$. We can derive that $r_j = x_j - t_j - s_j = x_j + s_j$ or $x_{j-1} + s_j$, then $-n + 2 \leq r_j \leq n$. Since $\eta = n$, $|r_j| \leq \eta$. In summary, $|r_j| \leq \eta$ if $u_j \in Y_i$ and x_j is repaired. Since L provide coverage for the set $U = \{u_1, \dots, u_n\}$, we can repair x_i for each $Y_i \in L$ to obtain a series that satisfies the seasonal and trend constraint. Therefore, $\Delta(x, x') = \text{card}(L) \leq \tau$.

Next we must demonstrate that any repair satisfying $\Delta(x, x') \leq \tau$ provides $\text{card}(L) \leq \tau$. Assume that there exists $\text{card}(L) > \tau$ when $\Delta(x, x') \leq \tau$. Note that if we only repair point x_i but $u_j \notin Y_i$, then $r_j = n + 1$ or $n + 1 + x_{i-1} > n$. Therefore, we need to repair x_i where $u_j \in Y_i$ to make r_j satisfy the seasonal and trend constraint. So $\Delta(x, x') \geq \text{card}(L)$ is a necessary condition if the entire series to be repaired, i.e., $\Delta(x, x') \geq \text{card}(L) > \tau$, which obviously contradict the initial assumption. In conclusion, we have demonstrated the NP-completeness of the decision problem at hand. $\square$

A.2 Proposition 1

PROOF. First, consider the trend filter

$$t = \text{median} \left(\left[x_{i+j}; \lfloor -\frac{m-1}{2} \rfloor \leq j \leq \lfloor \frac{m-1}{2} \rfloor \right] \right).$$

We define $A_k = [x_{k+l}; \lfloor -\frac{m-1}{2} \rfloor \leq l \leq \lfloor \frac{m-1}{2} \rfloor]$ as the trend estimation vector of the original series with violation x_i , where x_k is the k -th element of the series. Similarly, we define $B_k = [x_{k+l}; \lfloor -\frac{m-1}{2} \rfloor \leq l \leq \lfloor \frac{m-1}{2} \rfloor \wedge l \neq i, x_i^*]$ as the trend estimation vector of the clean series. Since we have $x_i^* \geq \max(x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)$, x_i^* is the largest number in B_k . Further, since $x_i > x_i^*$, x_i is also the largest number in A_k . Considering that the median is the value separating the higher half from the lower half of the data, and the order of the elements in A_k and B_k is the same, the median median A_k and median B_k is also the same, i.e., the trend component on $[x_1, \dots, x_i, \dots, x_j, \dots, x_n]$ with violations is exactly the same as that on the clean series $[x_1, \dots, x_i^*, \dots, x_j^*, \dots, x_n]$.

Then consider the seasonal filter

$$s_i = \text{median}(\{d_j; 1 \leq j \leq n \wedge (j - i) \mid m\})$$

We define $C_k = [d_j; 1 \leq j \leq n \wedge (j - k) \mid m]$ as the trend estimation vector of the original series with violation x_i , and $B_k = [d_j; 1 \leq j \leq n \wedge (j - k) \mid m \wedge l \neq i, x_i^*]$ as the trend estimation vector of the clean series. Similarly to the trend component, since $x_i^* \geq \max(x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)$, x_i^* is the largest number in D_k . Further, since $x_i > x_i^*$, x_i is also the largest number in C_k . Therefore, the order of the elements in C_k and D_k is the same. Then the median median C_k and median D_k is also the same, i.e., the seasonal component on $[x_1, \dots, x_i, \dots, x_j, \dots, x_n]$ with errors is exactly the same as that on the clean series $[x_1, \dots, x_i^*, \dots, x_j^*, \dots, x_n]$.

For the minimum value x_j , there is a similar conclusion. In summary, the violation-tolerant filter by Formulas 4 and 5 on $[x_1, \dots, x_i, \dots, x_j, \dots, x_n]$ with errors is exactly the same as that on the clean series $[x_1, \dots, x_i^*, \dots, x_j^*, \dots, x_n]$. $\square$

A.3 Proposition 2

PROOF. Let $W_k = \{x_l; ((l - k) \mid m) \wedge (l \neq k)\} \cup \{x_l; 1 \leq l \leq m\}$ for $1 \leq k \leq m$ in a original series, and $Y_k = W_k - \{x_k\} + \{x_k^*\}$ for $1 \leq k \leq m$ in a clean series. Since component extension $t_k = \text{median}(\{x_l; 1 - \lfloor -\frac{m-1}{2} \rfloor \leq l \leq k + \lfloor \frac{m-1}{2} \rfloor\} \cup \{t_{1+\lfloor \frac{m-1}{2} \rfloor} + s_l; m + k - \lfloor -\frac{m-1}{2} \rfloor \leq l < m + 1 - \lfloor -\frac{m-1}{2} \rfloor\})$, t_k is obtained by calculating the median of the elements in W_k multiple times. Since $x_i^* \geq \max(W_k)$, x_i^* is the largest number in Y_k . Further, since $x_i > x_i^*$, x_i is also the largest number in W_k . Considering that the median is the value separating the higher half from the lower half of the data, and the order of the elements in W_k and Y_k is the same, the median in W_k and Y_k is also the same, i.e., the trend extension on $[x_1, \dots, x_i, \dots, x_j, \dots, x_n]$ with errors is exactly the same as that on the clean series $[x_1, \dots, x_i^*, \dots, x_j^*, \dots, x_n]$. For the minimum value x_j , there is a similar conclusion. $\square$